

MTH:2310 DISCRETE MATH
QUIZ 6
DUE: FRIDAY OCT. 24

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. (14 points) *Solving Linear Recurrence Relations*

Suppose $(r-2)^3(2r-3)^2(3r-1)(r+1)(r+2)^2 = 0$ is the characteristic equation for a recurrence relation.

(a) (2 points) What is the degree of the recurrence relation?

$$3 + 2 + 1 + 1 + 2 = 9$$

Degree 9

(b) (4 points) Find all characteristic roots and state their multiplicity.

$$r_1 = 2, r_2 = \frac{3}{2}, r_3 = \frac{1}{3}, r_4 = -1, r_5 = -2$$

$$m_1 = 3, m_2 = 2, m_3 = 1, m_4 = 1, m_5 = 2$$

(c) (8 points) What is the general form of the solution to the recurrence relation?

$$a_n = (A + Bn + Cn^2) \left(\frac{3}{2}\right)^n + (D + En) \left(\frac{4}{3}\right)^n + F(-1)^n + (K + Ln)(-2)^n$$

Problem 2. (12 points) *Solving Linear Recurrence Relations*

Consider the recurrence relation $a_n = -9a_{n-1} - 27a_{n-2} - 27a_{n-3}$ with initial conditions $a_0 = -4$, $a_1 = 6$, $a_2 = 18$.

(a) (3 points) Find the characteristic roots associated to the recurrence relation. State their multiplicities.

$$r^3 + 9r^2 + 27r + 27 = 0 \quad \text{try } -3 \quad -27 + 81 - 81 + 27 = 0$$

$$(r+3)(r^2+6r+9) \quad r^2+6r+9 = (r+3)^2 \quad \text{thus our root is}$$

$$\rightarrow \text{w/ a multiplicity of } 3$$

(b) (4 points) State the general form of the solution to the recurrence relation.

$$a_n = (A + Bn + Cn^2) (-3)^n$$

Name _____

Section _____

(c) (5 points) Find the exact solution to the recurrence relation.

$$a_0 = (A + B(0) + C(0))(-3)^0 = -4 \quad \therefore A = -4$$

$$a_1 = (-4 + B + C)(-3)^1 = 6 \equiv B + C = 2$$

$$a_2 = (-4 + 2B + 4C)(-3)^2 = 18 \equiv -36 + 18B + 36C = 18$$

$$A = -4$$

$$B + C = 2 \equiv B = 2 - C$$

$$18B + 36C = 54 \equiv B + 2C = 3$$

$$2 - C + 2C = 3$$

$$C = 1$$

$$B + 1 = 2 \quad B = 1$$

$$A = -4 \quad B, C = 1$$

$$a_n = (-4 + n + n^2)(-3)^n$$

Problem 3. (12 points) Mathematical Induction

Prove that for every positive integer n , $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n \cdot (n+1)} =$

$$\frac{n}{n+1} \cdot \quad P(n) = \sum_{j=1}^n \frac{1}{j(j+1)} = \frac{n}{n+1}$$

Base case: $P(1) = \sum_{j=1}^1 \frac{1}{j(j+1)} = \frac{1}{1+1} = \frac{1}{2} = \frac{1}{2}$ thus the base

case is true

assume there exists $k \in \mathbb{Z}^+$ and $k \geq n_0$ such that $P(k)$ is true. we now prove $P(k+1)$

$$\sum_{j=1}^{k+1} \frac{1}{j(j+1)} = \frac{k+1}{k+2}$$

the final term in the LHS summation is $\frac{1}{(k+1)(k+2)}$

therefore we can rewrite as $\frac{1}{(k+1)(k+2)} + \sum_{j=1}^k \frac{1}{j(j+1)} = \frac{k+1}{k+2}$

from $P(k)$, we know $\sum_{j=1}^k \frac{1}{j(j+1)} = \frac{k}{k+1}$

$$\frac{k+2}{k+2} \left(\frac{1}{(k+1)(k+2)} + \frac{k}{k+1} \right) = \left(\frac{k+1}{k+2} \right) \frac{k+2}{k+2}$$

$$\frac{1}{(k+1)(k+2)} + \frac{(k+2)k}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

$$\frac{(k+2)k+1}{(k+1)(k+2)} = \frac{k+1}{k+2} = \frac{k^2+2k+1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

thus we show by induction that $\sum_{j=1}^n \frac{1}{j(j+1)} = \frac{n}{n+1}$ is true where $n \in \mathbb{Z}^+$

Problem 4. (12 points) Mathematical Induction

Prove that for any non-negative integer n , 6 divides $7^n - 1$. Recall that a non-zero integer b divides an integer a , if there is an integer q such that $b = qa$.

$$P(n) = 6 \mid (7^n - 1)$$

$$\text{Let } n=0, \quad P(0) = 6 \mid (7^0 - 1) = 6 \mid (1 - 1) = 6 \mid 0 \equiv T$$

thus, our base case is true.

assume that $P(k)$ is true, where $k \in \mathbb{Z}^+ \wedge k \geq 0$.

$$P(k) = 6 \mid 7^k - 1 \equiv 7^k - 1 = 6m \text{ where } m \in \mathbb{Z}$$

$$P(k+1) = 6 \mid (7^{k+1} - 1)$$

for this to be true, there must exist $q \in \mathbb{Z}$ such that

$$6q = 7^{k+1} - 1 \quad 7^{k+1} - 1 = 7(7^k) - 1$$

$$= 7(7^k) - 7 + 7 - 1 = 7(7^k - 1) + 6 = 7(6m) + 6 = 6(7m + 1)$$

as $m \in \mathbb{Z}$, $(7m + 1) \in \mathbb{Z}$ thus there exists an integer q for which

$$6q = 7^{k+1} - 1 \text{ is true}$$

therefore, we've proven $6 \mid 7^n - 1$ for $n \in \mathbb{N}$ by induction.

